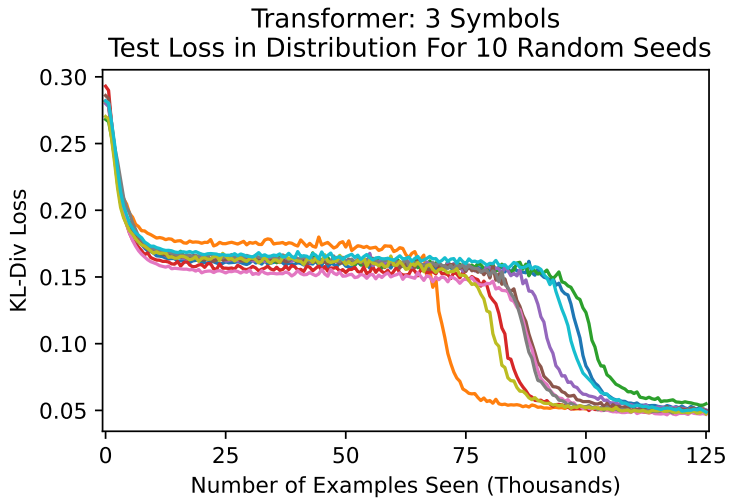

Figure 8: In distribution test loss for 10 two layer attention only transformers, with random seeds $0, 1, \ldots 9$ (randomness affects initialization and the training data). The training dynamics are consistent for each model, though the exact position of the phase transitions.

where δ is Dirac's delta function.

Thus, after 2 steps of gradient descent, as $t \to \infty$, the prediction of the model will be

$$f(e)_{p,i} = O(\epsilon^4) \sum_{t'=1}^{p} \mathbb{1}\{x_{t'} = i\} \mathbb{1}\{x_p = x_{t'-1}\} + O(\epsilon^6). \tag{A.68}$$

□

An interesting observation that stems from the proof is that the learning rate of the first step is $O(1/t^2)$, where recall t is the sequence length, while for the second step the learning rate is $O(1/t)$ (much larger). The first step in our proof corresponds to the learning of the second layer W_k, while the second step "cleans up" W_k and learns the first layer v. Interestingly, this is what we also observe in the experiments with the transformers and this minimal model: the first drop is immediate for the learning of the 2nd layer, while the second loss drop happens after a long plateau and corresponds to the "grokking" of the positional embeddings.

B Experimental Details

We train transformers of the form (4) with the AdamW optimizer with learning rate $3e-5$, batch size 64, and hidden dimension 16. The sequence length of the examples is 100 tokens. The minimal model was trained with SGD, with batch size 64, and learning rate $2e-3$. For 3-grams, a learning rate of $3e-2$ was used. We use PyTorch 2.1.2. Some of the training and model code was based on minGPT Karpathy (2023).

The experiments all measure the outputs of the models at the last token.

Distributions Used in Figure 4a This graph is for $k = 2$ symbols, to simplify the distributions used. The exact method of generating the distributions used in Figure 4a is as follows. Let $p \in (0, 1)$ parametrize the distribution. Choose a uniformly random number $x \sim U(0, 1)$, then define $\mu = x + p(1 - 2x)$. Then choose $y \sim U(\mu - 0.2, \mu + 0.2)$, and then $y = max(0, min(1, y))$. Then, with probability $1/2$, $a = x$ and $b = y$,

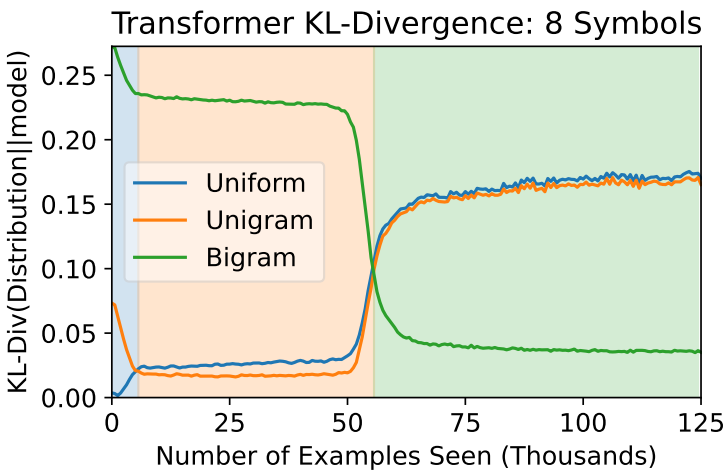

Figure 9: Our results extend to more symbols than $k = 2$ or $k = 3$. The KL-divergence between the transformer and strategies over training. This required a sequence length greater than 100 (200 in this case) for the difference between unigrams and bigrams to be large enough for the unigram phase to be visible (in either case there was a plateau before the final drop in test loss).

otherwise $a = y$ and $b = x$. Then the transition matrix is

$$\begin{pmatrix} a & 1-a \\ b & 1-b \end{pmatrix}.$$

C Additional Experiments

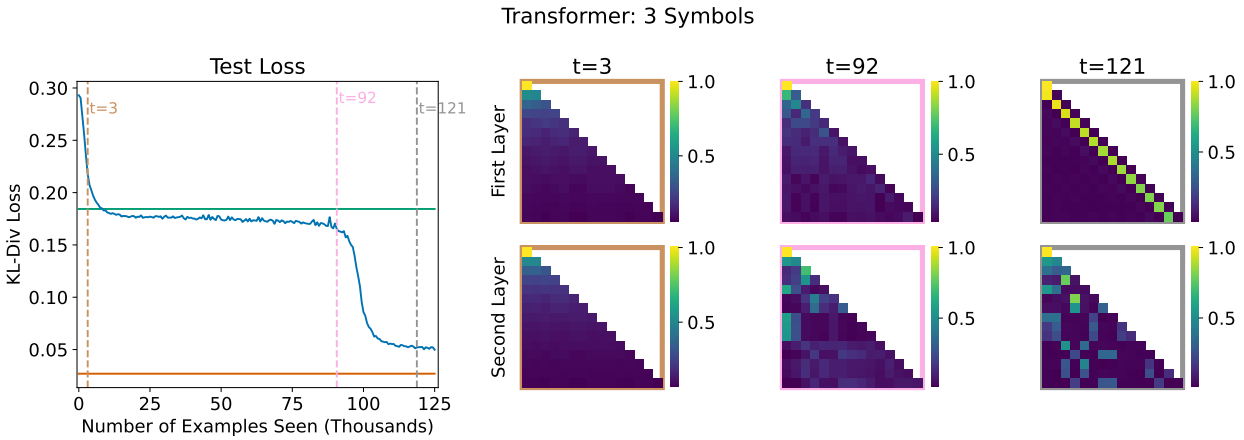

Figure 10: A two layer attention-only transformer was trained with cross entropy loss on ICL-MC. The heatmaps on the right represent part of the attention for the transformer at various time steps, specifically the value of A from (5). The top row are showing A from the first layer, and the bottom row from the second layer.

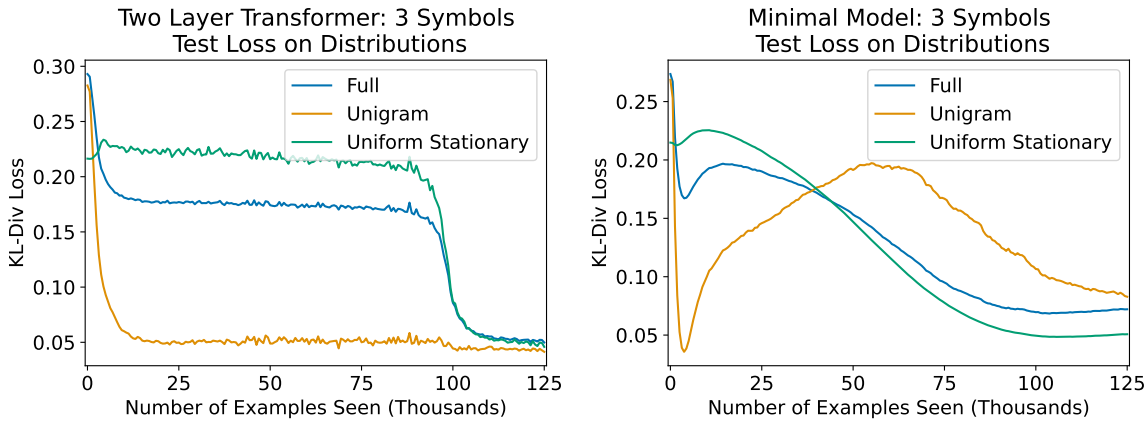

Figure 11: A two layer attention-only transformer (top) and minimal model (8) (bottom), trained on the main task with ICL-MC with cross entropy loss, test loss measured by KL-Divergence from the underlying truth (labels based on transition probabilities, not samples). The distributions test loss is measured in are (from left to right) in-distribution, a distribution where each token is sampled iid, and a distribution over uniformly random doubly stochastic transition matrices (equivalently, stationary distribution is identity, or unigram based guesses are as good as guessing uniform probability). For both models, the in distribution test loss quickly drops to the level of the unigram algorithm.

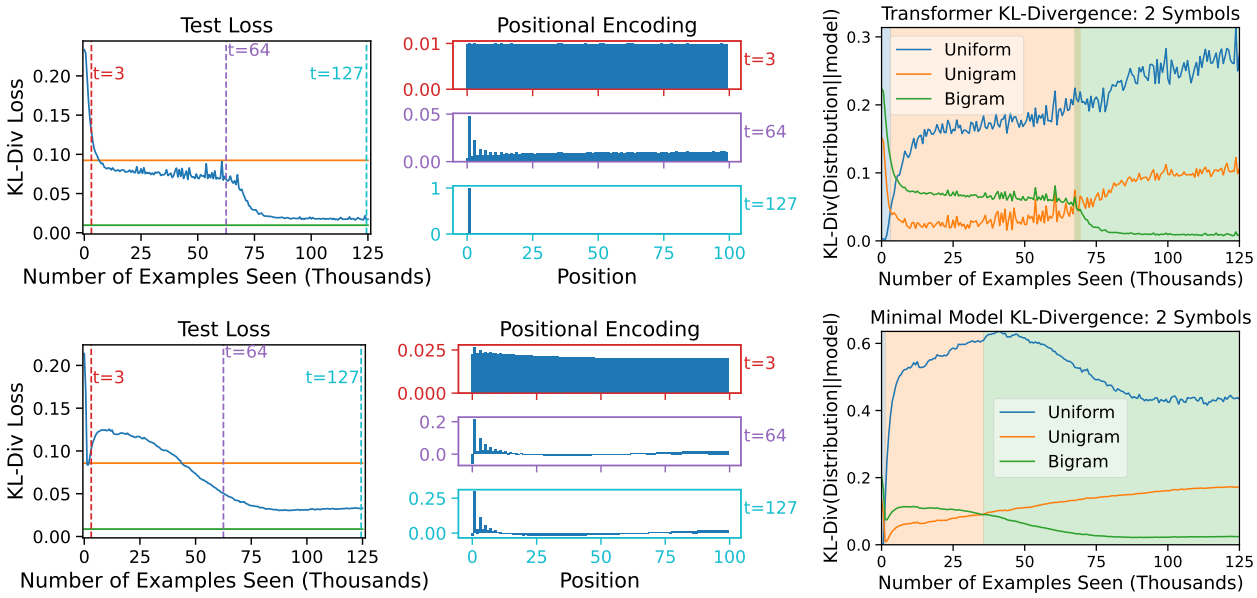

Figure 12: A comparison of the two layer attention only transformer and minimal model for $k = 2$ symbols. Note the alternating pattern in the positional encodings at 64,000 examples seen.